

TOPIC 01 · ADVANCED DATA SCIENCE PROJECT · 5DS

BI Flow

Multi-Agent Automation of the Power BI Pipeline

Raw trade data in. Explained dashboards out.

Project report — Design Thinking approach and BO / DSO mapping
Business case: import / export trade data

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Abstract

Business intelligence (BI) teams still build most of their reports by hand. In a typical Power BI workflow, an analyst imports the data, cleans it, models it, writes DAX measures and designs the visuals, and repeats these steps for every new report. The work is slow, error-prone and hard to audit.

BI Flow is a multi-agent system that automates this pipeline. Each Power BI step is reproduced by a specialised AI agent (profiling, cleaning, ETL, KPI definition, analysis, dashboard generation and explainability), and an orchestrator agent coordinates them. The system is tested on import / export trade data, a domain where files come from many sources, mix currencies and product codes, and must remain traceable for customs audits.

This report follows the six steps of the Design Thinking method used in the module (Empathise, Define, Ideate, Prototype, Test, Implement), then maps the project's six Business Objectives (BO) to eight Data Science Objectives (DSO) and to the agents that deliver them.

Keywords: business intelligence, Power BI, multi-agent systems, agentic AI, LLM, explainable AI (XAI), data quality, Design Thinking, import / export.

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CHAPTER 1

Introduction

1.1 General context

Three trends frame this project.

Business intelligence Organisations rely on BI tools such as Power BI to turn raw data into decisions. Yet every new report goes through the same five manual steps: *Get Data* → *Power Query* → *Data model* → *DAX measures* → *Visuals*. The analyst's time goes mostly into preparation rather than analysis.

Agentic AI Language-model agents can now plan a task, call tools and pass results to each other. A multi-agent system splits a complex job into specialised roles under one orchestrator, which makes each role easier to test, explain and improve.

Business case: import / export trade data Trade files mix customs declarations, ERP exports and spreadsheets, several currencies, inconsistent country names and HS product codes. Every figure must also hold up in a customs audit, which makes this domain a demanding test for automated BI.

1.2 Project goal

Design a multi-agent architecture that automates the full BI pipeline: from a raw dataset to an intelligent, explainable and automated dashboard.

1.3 Scope: what the system must do

Table 1.1 lists the five capabilities the system must deliver and the agents responsible for each.

Table 1.1: Required capabilities and responsible agents.

#	Capability	Agent(s)
01	Analyse heterogeneous data automatically	Data Profiler
02	Clean and transform the data	Quality & Cleaning, ETL
03	Compute the relevant KPIs	Semantic & KPI
04	Generate visualisations and insights	BI Analyst, Dashboard Generator
05	Explain results transparently (XAI)	BI Auditor / XAI

The architecture is designed to work across sectors (e-commerce and retail, banking and finance, telecommunications, transport and tourism, health, energy and utilities). The team starts with one

business domain: **import / export trade**.

1.4 Methodology: Design Thinking

The project follows the Design Thinking method required by the module. Design Thinking starts from the users rather than from the technology, and runs as a loop: feedback gathered after deployment starts a new cycle (Figure 1.1).

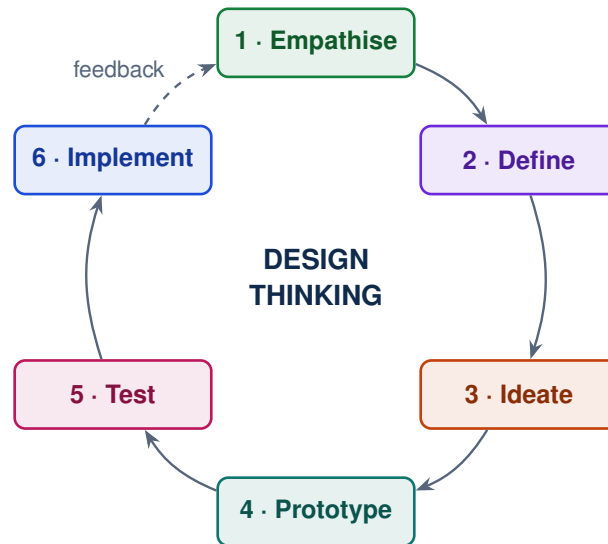


Figure 1.1: The six-step Design Thinking cycle followed in this project.

The following chapters apply each step to BI Flow. Chapter 8 then maps the business objectives to data science objectives and agents.

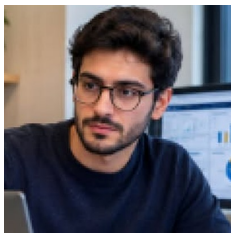
CHAPTER 2

Empathise — understanding the users

The first step is to understand the people who live the problem, before proposing any solution. Three personas represent the main users of an import / export BI system.

Note The personas below are working hypotheses built from the team's understanding of the domain. They will be validated through interviews with practitioners (BI analysts, export managers and compliance officers).

Persona 1 — Karim Trabelsi, BI Analyst



Age 29 · **Status** employee · **City** Tunis

"I'm tired of redoing the same data cleaning every week instead of actually analyzing."

Bio. Manages the company's BI reports daily. Skilled in Excel, SQL and visualisation tools, but much of his time goes into manual cleaning and fixing DAX formula errors.

Needs. Automate repetitive data preparation. **Expectations.** A reliable pipeline that frees time for high-value analysis.

Frustrations. More time on cleaning than analysis; fear of DAX errors that silently distort reports.

Persona 2 — Leila Ben Ahmed, Export Director



Age 41 · **Status** senior executive · **City** Tunis

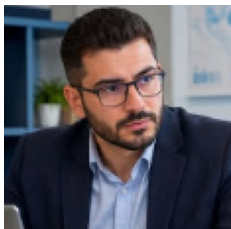
"I need to be able to tell my clients where things stand without waiting three days for a report."

Bio. Oversees export activity across several continents, juggling markets, B2B clients and sales teams. Needs consolidated data to decide fast.

Needs. Fast, reliable country / product KPIs. **Expectations.** A dashboard she can check at any time, without waiting for a manual report.

Frustrations. Late reports; contradictory figures between teams that undermine her credibility.

Persona 3 — Sami, Customs Compliance Manager



Age 35 · **Status** employee · **City** Tunis

"If there's an audit, I need to be able to justify every number, not just trust it."

Bio. Ensures customs compliance for import / export operations and regularly justifies figures to the authorities.

Needs. Know where every figure comes from. **Expectations.** Full traceability of data transformations, from source to final KPI.

Frustrations. Corrections made without traceability; fear of an audit where he cannot justify a figure.

2.1 Key insights

1. **Data preparation dominates the analyst's time**, at the expense of analysis.
2. **The same KPIs are recalculated for every report**, by hand.
3. **Without explanations, decision makers do not trust the figures**, and compliance cannot defend them.

Define — the problem statement

3.1 Point of view and “How might we” question

Import / export analysts lose most of their time on repetitive Power BI steps, which produces slow, unreliable and unexplainable reports for decision makers. The team turned this point of view into one guiding question:

HOW MIGHT WE...

...automatically transform raw import / export data into a reliable and explained Power BI dashboard?

Each word answers one persona: *automatically* for Karim, *reliable* for Leila, *explained* for Sami.

3.2 The problem in numbers

Three published figures show that the problem is widespread. They describe data work in general, not import / export specifically, and are used here as context.

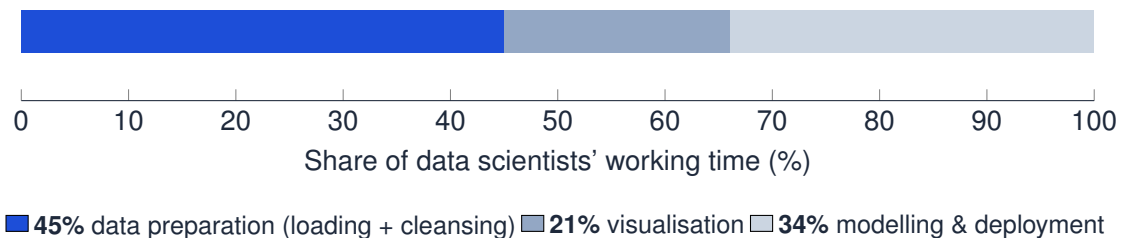


Figure 3.1: Time. About 45% of data scientists' time goes into loading and cleaning data [1]. This is Karim's pain.

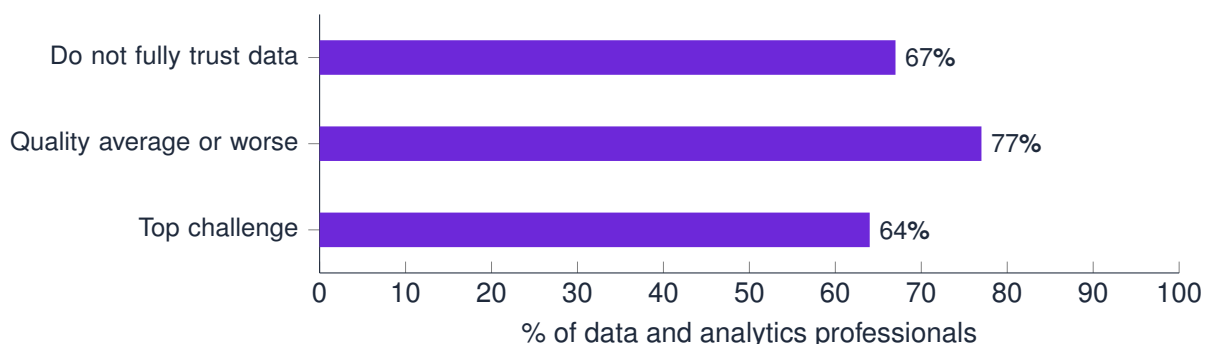


Figure 3.2: Trust. 67% do not fully trust the data their organisation uses for decisions; 77% rate their data quality average or worse; 64% name data quality their top data-integrity challenge (565 respondents, 2024) [2]. This is Leila's pain.

Cost Poor data quality costs organisations **at least \$12.9 million a year on average** [3], roughly \$35,000 a day. For Sami, every untraced correction also adds audit risk.

Ideate — generating and choosing a solution

4.1 Four solution ideas

The team brainstormed four ways to answer the “How might we” question (Table 4.1). Options A to C are conventional approaches; option D is the multi-agent architecture.

Table 4.1: Solution ideas with their main strength and limit.

Idea	Strength	Limit
A · Power BI alone (reusable .pbit template)	Simple and fast to set up	Rigid: breaks when the data structure changes
B · Fixed ETL scripts	Fast execution	No intelligence, heavy maintenance
C · Single LLM chatbot	Flexible, natural language	Hallucinations, not traceable
D · Multi-agents + orchestrator	Modular, reliable, explainable	More complex coordination

4.2 Evaluation against the personas’ needs

Each idea was scored against four criteria derived from the personas (Table 4.2 and Figure 4.1).

Table 4.2: Evaluation matrix (A: Power BI alone, B: fixed scripts, C: LLM chatbot, D: multi-agents).

Criterion	A	B	C	D
Speed (Karim)	Yes	Yes	Yes	Yes
Reliability (Leila)	Yes	Yes	No	Yes
Traceability (Sami)	No	No	No	Yes
Adaptability	No	No	Yes	Yes
Score	2/4	2/4	2/4	4/4

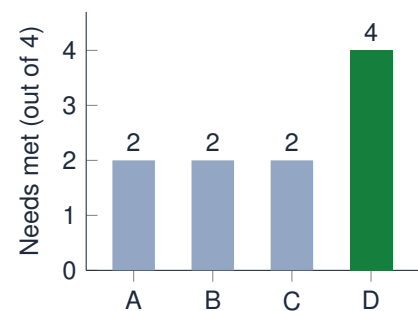


Figure 4.1: Scores of the four ideas.

Only D meets all four needs; traceability, Sami’s need, is what eliminates A, B and C.

Selected approach: D. Each agent reproduces one Power BI step and can be tested on its own; the orchestrator keeps the pipeline consistent; the explainability agent answers the traceability need.

Prototype — the multi-agent architecture

5.1 Architecture

Figure 5.1 shows the architecture of the prototype. Raw import / export files enter the pipeline, flow through seven specialised agents and come out as a dashboard with its reports. The orchestrator is linked to every agent; a cross-cutting layer provides shared memory, logs, human validation and a retrieval-augmented (RAG) trade glossary.

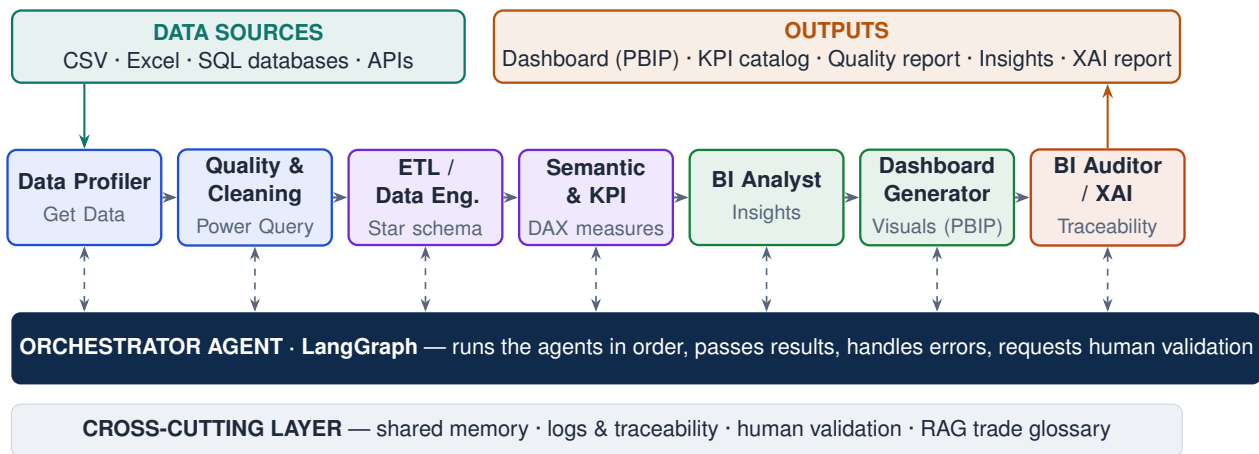


Figure 5.1: Multi-agent architecture of BI Flow: one agent per Power BI step, coordinated by an orchestrator. Dashed arrows are orchestration links.

5.2 Technology stack

- **Agents and orchestration:** Python, LangGraph, large language models (LLM).
- **Knowledge:** retrieval-augmented generation (RAG) over a trade glossary (vector store, e.g. ChromaDB).
- **Data:** pandas and SQL for profiling, cleaning and modelling.
- **BI output:** Power BI project format (PBIP), whose model and report are stored as text files that an agent can generate.
- **Engineering:** GitHub for versioning and CI, Docker for reproducible execution.

5.3 Agent roles

Table 5.1: Role of each agent, the Power BI step it replaces and an import / export example.

Agent	Replaces in Power BI	Import / export example
Orchestrator	Coordination	Runs agents in order, handles errors, asks for validation
Data Profiler	Get Data	Detects types, duplicates, missing countries and invalid values
Quality & Cleaning	Power Query	Normalises countries (ISO codes), HS codes and currencies
ETL / Data Engineering	Data modeling	Builds Fact_Trade with Country, Product and Date dimensions
Semantic & KPI	DAX measures	Trade balance, coverage ratio, year-over-year growth
BI Analyst	Analysis	Trends, anomalies, forecasts and recommendations
Dashboard Generator	Visuals	Picks the charts and generates the Power BI report (PBIP)
BI Auditor / XAI	Governance	Explains every correction, KPI and insight, with its source

CHAPTER 6

Test — validating the prototype

Testing checks two things: that the prototype **works technically** and that it is **useful to its users**.

6.1 Technical tests (per agent)

Table 6.1: Technical tests.

Agent	What is checked	Method
Profiler, Cleaning	Data quality	Before vs after comparison
ETL	Model output	Scripts run without error
Semantic & KPI	KPI accuracy	Comparison with a manual Power BI report
Analyst, Dashboard	Relevance	Evaluation grid
Orchestrator	Robustness	Timing and success rate of full runs

6.2 User tests (personas)

- **Karim:** does he really save time?
- **Leila:** can she read the dashboard without help?
- **Sami:** can he trace the origin of any figure?

The explainability agent must also answer four questions: *Why was this value corrected? Why this KPI? Where does the insight come from? How is the recommendation justified?*

6.3 Decision rule



Figure 6.1: A failed test sends the team back to the prototype; a passed test moves the agent to implementation.

Phase 4 deliverable: notebook, test report and oral presentation.

Implement — deployment and deliverables

7.1 Deployment chain

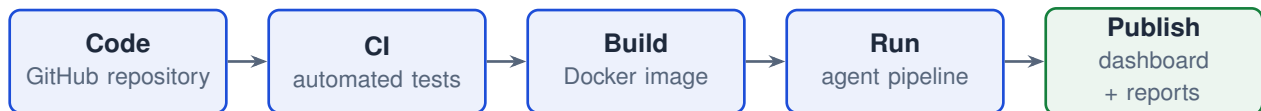


Figure 7.1: Deployment chain: every change is tested, packaged and run end to end.

7.2 Final deliverables

- Architecture document
- Source code (GitHub)
- Power BI dashboard
- KPI catalog
- Data quality report
- Insights and recommendations
- XAI (explainability) report
- Final defense

Real user feedback after deployment feeds a new Empathise step, closing the Design Thinking loop.

Business Objectives and Data Science Objectives

Each Business Objective (BO) is translated into one or two Data Science Objectives (DSO), and each DSO is delivered by a specialised agent. All agents are supervised by the Orchestrator Agent.



Figure 8.1: From the business need to explainable BI outputs.

Table 8.1: BO / DSO mapping and responsible agents.

Business objective	Data science objective	Agent
BO1 — Improve data management and organisation: optimise structure and governance for more accessible data.	DSO1 — Automatic data profiling: identify structure, characteristics and quality issues.	Data Profiler
BO2 — Improve data quality: guarantee reliable, consistent and usable data.	DSO2 — Data quality & cleaning: detect and fix quality issues to produce a reliable dataset. DSO3 — Transformation & ETL: turn cleaned data into a coherent analytical model.	Data Quality & Cleaning ETL / Data Engineering
BO3 — Improve visibility on performance: a consolidated view of commercial and logistics performance.	DSO4 — KPI & business semantics: define and compute business indicators automatically.	BI Semantic & KPI
BO4 — Improve customer profitability analysis: rank B2B customers by their real contribution to profit.	DSO5 — Automatic analysis: detect trends, variations, anomalies and related factors.	BI Analyst
BO5 — Optimise logistics performance: analyse costs, lead times, routes and carriers.	DSO6 — Insights & recommendations: turn analyses into actionable recommendations.	BI Analyst
BO6 — Enable transparent decision-making: help decision makers see where KPIs, insights and recommendations come from.	DSO7 — Dashboard generation: visuals suited to the KPIs and analyses. DSO8 — Explainability & audit: traceable explanations of every result.	Dashboard Generator BI Auditor / XAI

BO1 to BO3 are the foundations (organised data, quality data, visible performance). BO4 to BO6 build on them (analysis, recommendations and transparent decisions). BO2 and BO6 each need two data science objectives.

Conclusion and next steps

9.1 What BI Flow brings

- **Faster (Karim):** the report that took days is produced by the agent pipeline.
- **More reliable (Leila):** profiling and cleaning agents keep country codes, HS codes and currencies consistent.
- **Explainable (Sami):** every correction, KPI and recommendation is justified and traceable for audits.

Key message. Design Thinking turned a real user problem into a multi-agent architecture where one agent reproduces one Power BI step, from raw import / export files to an explained dashboard.

9.2 Next steps

1. Validate the three personas through interviews with practitioners.
2. Select and document the public import / export dataset used for the prototype.
3. Build the first agents (profiling and cleaning), then the model, KPI and orchestration layers.
4. Benchmark the generated dashboard against a manual Power BI report (accuracy and time).
5. Extend the pipeline to other business sectors.

9.3 Points of attention

- The automatic generation of the Power BI project (PBIP) must be validated technically.
- Corrections proposed by language models must remain under human control.
- Data ethics, bias and governance will be documented in the final phase.

References

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